

Histone-Modifier Graph Transformer for Prognosis Prediction in Hepatocellular Carcinoma

*Thesis submitted to the Amrita Vishwa Vidyapeetham in partial fulfilment of the
requirement for the Degree of*

MASTER of TECHNOLOGY

in

Artificial Intelligence

Submitted by

Surya Gayathri S AM.SC.P2ARI24014



**AMRITA SCHOOL OF COMPUTING
AMRITA VISHWA VIDYAPEETHAM
(Estd. U/S 3 of the UGC Act 1956)
AMRITAPURI CAMPUS**

KOLLAM -690525

June 2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

AMRITA VISHWA VIDYAPEETHAM

(Estd. U/S 3 of the UGC Act 1956)

Amritapuri Campus

Kollam -690525



BONAFIDE CERTIFICATE

This is to certify that the thesis entitled "**Histone-Modifier Graph Transformer for Prognosis Prediction in Hepatocellular Carcinoma**" submitted by **Surya Gayathri S (AM.SC.P2ARI24014)**, in partial fulfillment of the requirements for the award of Degree of Master of Technology in Artificial Intelligence from Amrita Vishwa Vidyapeetham, is a bonafide record of the work carried out by him/her under my guidance and supervision at Amrita School of Computing, Amritapuri during Semester 4 of the academic year 2025-2026.

Dr. Geetha M

Thesis Advisor

Dissertation Coordinator

Dr. Swaminathan J

Chairperson

Dept. of Computer Science & Engineering

Amrita School of Computing

Reviewer

Place : Amritapuri

Date : 18/06/2026

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

AMRITA VISHWA VIDYAPEETHAM

(Estd. U/S 3 of the UGC Act 1956)

Amritapuri Campus

Kollam -690525



DECLARATION

I, **Surya Gayathri S(AM.SC.P2ARI24014)** hereby declare that this project entitled "**Histone-Modifier Graph Transformer for Prognosis Prediction in Hepatocellular Carcinoma**" is a record of the original work done by me under the guidance of **Dr.Geetha M**, Dept. of Computer Science and Engineering, Amrita School of Computing, Amrita Vishwa Vidyapeetham, that this work has not formed the basis for any degree/diploma/associationship/fellowship or similar awards to any candidate in any university to the best of our knowledge.

Place : Amritapuri

Date : 18/06/2026

Signature of the Student

Signature of the Thesis Advisor

Acknowledgements

I would like to express my sincere gratitude to my thesis advisor, Dr. Geetha M, for the sustained guidance, constructive criticism, and intellectual encouragement provided throughout this work. The clarity of perspective brought to the biological and computational dimensions of the problem shaped this project in ways that go well beyond what any student could achieve independently.

I am grateful to the faculty of the Department of Computer Science and Engineering at Amrita School of Computing, Amritapuri, for creating a research environment in which questions about the intersection of computation and biology can be explored with rigour and curiosity. The feedback received during seminars and review sessions was invaluable.

Abstract

Hepatocellular carcinoma (HCC) is the third most common cause of cancer-related deaths globally and continues to be one of the deadliest types of cancer. Many HCC patients still have a dismal prognosis despite tremendous advancements in immunotherapy, targeted treatments, and surgical techniques. One of the main issues is that traditional staging methods frequently fail to account for the tumor’s underlying molecular complexity because they mainly rely on the tumor’s clinical and morphological features. Because of unique genetic and epigenetic changes that affect tumor behavior, development, and response to treatment, patients categorized within the same clinical stage may have noticeably diverse results.

This thesis presents HisGT-HCC, a Histone-modifier Graph Transformer for predicting HCC prognosis. The main premise is that histone-modifying enzymes are part of interacting protein complexes and that the coordinated network dysregulation of these complexes, as opposed to the individual expression of a complex member, represents the epigenetic phenotype of an HCC tumour. To view the network, a protein-protein interaction (PPI) graph was built for 114 histone-modifier genes, spanning ten enzyme families (HAT, HDAC, HMT, KDM, PRMT, SIRT, DNMT, TET, CBX and BRD). A confidence score of 700 was used in both the EpiFactors database and the

STRING PPI database. Compared to a baseline graph, which drew from four of the ten families and had 94 nodes and 736 edges, the extended 10-family graph included 313 nodes and 4,278 edges.

TCGA-LIHC RNA-seq data from 365 patients were used to apply a LASSO-penalised Cox proportional hazards model to the hub genes of this network. 56 prognostic genes were selected to construct a composite risk signature. A Graph Transformer with two graph attention layers was then trained on a patient similarity graph over the LASSO-selected features, optimised with the Cox partial likelihood loss.

The model developed using TCGA-LIHC cohort achieved C-index of 0.748. This is a considerable improvement from the baseline of 0.644 achieved using the baseline four-family model. The high and low-risk groups (log-rank $p = 4.34 \times 10^{-16}$) shows a clear and significant distinction in survival through Kaplan-Meier analysis. Multivariate Cox analysis shows that the LASSO composite risk score is significant, independent of patient age and the stage of the tumor (HR = 1.273, 95% CI: 1.078–1.504, $p = 0.004$). It is also significant that risk stratification is sustained in all seven clinical sub groups that were analyzed, including those patients considered to be in the early stage (Stage I/II, log-rank $p = 1.83 \times 10^{-9}$) of the disease, where the current clinical staging of the tumor fails to offer any prognostic value. In support of the biological validation of the signature, EZH2, HDAC2, PRMT5 and HDAC1, the genes which confer the most risk that were identified, are all known to be epigenetic regulatory oncogenes of liver cancer.

These findings indicate that utilizing graph structures to model histone-modifier interaction networks can reveal subpopulations of HCC that may largely go unrecognized using standard histological substaging methods. This has implications for the design of future clinical trials using epigenetic therapies, especially for treatment strategies targeting early-stage disease.

Keywords: hepatocellular carcinoma, histone modification, graph transformer, LASSO-Cox, protein-protein interaction, epigenetic prognosis, survival analysis

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Chapter 1

Introduction

Liver cancer remains one of the most lethal malignancies encountered in clinical practice, with hepatocellular carcinoma (HCC) accounting for nearly 75–85% of all primary liver cancer cases worldwide. According to the Global Cancer Statistics 2020 report, HCC ranked as the sixth most commonly diagnosed cancer and the third leading cause of cancer-related mortality, claiming over 830,000 lives annually [1]. The burden of this disease falls disproportionately on regions with high prevalence of hepatitis B virus (HBV) and hepatitis C virus (HCV) infection, including South-East Asia, sub-Saharan Africa, and parts of South America. Despite decades of research, the five-year overall survival rate for HCC remains below 20% in most clinical settings, largely because the majority of patients present at an advanced stage when curative options such as surgical resection, liver transplantation, or radiofrequency ablation are no longer feasible [2].

The poor outlook for HCC comes from two connected challenges. The first is biological. HCC develops through a complex, multi-step process of liver cancer growth driven by genomic instability, unusual signaling pathway activation, and widespread changes in gene expression. The second challenge is clinical.

Traditional staging systems, like the Barcelona Clinic Liver Cancer (BCLC) classification and the TNM system, only address basic anatomical and physiological features. They offer limited insight into the molecular biology of each tumor. As a result, two patients with the same stage can have very different survival rates. This disconnect between clinical details and molecular reality explains why staging alone often fails to inform personalized treatment choices. [3]

1.1 Motivation

In the molecular changes seen in HCC, disruption of the epigenetic control of gene expression has been given greater recognition. Chromatin modifiers – including writers responsible for adding marks to chromatin, erasers that erase those marks, and readers who understand those marks – control the transcription programs which define cellular function, growth, and survival. Imbalances among these enzymes lead to the constitutive activation of oncogenes that had been silenced, as well as to the permanent silencing of tumor suppressor genes.

Histone-modifying enzymes do not act alone. They work as part of multi-protein complexes, such as the NuRD complex with HDAC1 and HDAC2, the PRC2 complex with EZH2, and the COMPASS complex with KMT2A and SETD1A. The activity of any single enzyme depends on its interactions with others. This protein-protein interaction (PPI) network shows that measuring the expression of one gene does not give a complete view of a tumour’s epigenetic state. Instead, it is the coordinated activity across the entire histone-modifier network that shapes the epigenetic phenotype. This idea drives the current work.

While individual histone-modifying genes like EZH2, HDAC2, and KDM1A have been studied as predictors in HCC, no previous research has systematically

modeled the combined prognostic effect of the histone-modifier protein-protein interaction network using graph-based deep learning. Current computational methods treat each gene as a separate factor and overlook the structural details within the biological network.

The main question this thesis addresses is: can a graph-structured representation of histone-modifier protein interactions, along with patient-level gene expression data and a Graph Transformer trained on survival outcomes, create a prognostic model for HCC that is both biologically interpretable and clinically relevant?

1.2 Objectives

The specific objectives of this work are as follows:

1. Create a detailed PPI graph for 10 functional families of histone-modifying enzymes (HAT, HDAC, HMT, KDM, PRMT, SIRT, DNMT, TET, CBX, BRD) from the EpiFactors database and STRING PPI resource.
2. To get the TCGA-LIHC gene expression profiles for the extracted genes and created normalised adjacent and provide feature matrices for downstream graph learning.
3. To find hub genes and inter-family interactions. modeling for the identification of patterns in the PPI network, and for LASSO-Cox modeling. regression to build a composite prognostic risk score.
4. To build and train a Graph Transformer model using a graph. Similarity graph of patients based on Lasso-selected genes The expression features are optimised using Cox partial. likelihood loss.

5. To compare the model using the Kaplan-Meier test, A linear regression model was used for computing the concordance index, as well as clinical association. The analyses comprised multivariate Cox regression across. All major clinical and full TCGA-LIHC cohort. subgroups.
6. To give a biological interpretation of the identified gene signature in terms of known epigenetic mechanisms in HCC.

1.3 Thesis Structure

This thesis is organized into six chapters, each addressing a specific component of the study.

Chapter 1 provides an introduction to the problem domain, including the motivation behind the work, the research objectives, and an overview of the thesis organization.

Chapter 2 reviews the relevant literature across three areas: the epidemiology and clinical management of HCC, the role of histone modifications in cancer biology, and computational approaches to cancer prognosis using graph neural networks.

Chapter 3 formally defines the problem, introduces the mathematical notation used throughout, and identifies the key challenges that motivate the proposed approach.

Chapter 4 presents the proposed HisGT-HCC system in full—covering data sources, PPI graph construction, hub gene identification, LASSO-Cox signature derivation, patient graph building, and the Graph Transformer architecture and

training procedure.

Chapter 5 presents and discusses all experimental results, including baseline performance, expanded model performance, clinical association analysis, subgroup survival analysis, biological interpretation, and the external validation attempt on GSE14520.

Chapter 6 summarises the contributions of the thesis, acknowledges its limitations, and proposes directions for future research. Two appendices provide the complete 56-gene LASSO-Cox signature and the full inter-family interaction statistics.

Chapter 2

Related Work

Hepatocellular carcinoma develops almost exclusively on the background of chronic liver disease. The most common aetiological factors are chronic HBV infection, accounting for approximately 50% of cases globally, and HCV infection, contributing around 25%. Additional risk factors include alcoholic cirrhosis, non-alcoholic fatty liver disease (NAFLD), dietary exposure to aflatoxin B1, and rare hereditary conditions such as haemochromatosis. The transition from chronic hepatitis to cirrhosis to HCC typically takes two to three decades and involves progressive accumulation of genetic and epigenetic aberrations. [4]

Current staging systems attempt to stratify patients by disease burden, liver function reserve, and performance status. The BCLC system, widely adopted in Europe and North America, classifies patients into five stages (0, A, B, C, D) based on tumour number and size, portal vein invasion, extrahepatic spread, and Child-Pugh class. The AJCC/TNM system provides a complementary anatomical classification. While these staging systems have clear clinical utility, they are based on macroscopic parameters that do not capture the underlying molecular heterogeneity of the tumour. Two patients with identically staged HCC may

differ substantially in their epigenetic landscape, driving differences in biological aggressiveness that staging alone cannot capture [5].

2.1 Existing Approaches

2.1.1 Epigenetic Dysregulation in Hepatocellular Carcinoma

Global DNA hypomethylation, promoter-specific hypermethylation of tumor suppressor genes, and extensive changes in histone modification patterns are characteristics of the epigenome of HCC. A significant percentage of HCC cases exhibit overexpression of EZH2, the catalytic subunit of the PRC2 complex in charge of H3K27 trimethylation, which is linked to a poor prognosis. In addition to promoting the epithelial-to-mesenchymal transition that aids in metastasis, EZH2 silences a number of tumor suppressor genes. [6] Members of the NuRD complex, HDAC1 and HDAC2, are often overexpressed in HCC. They deacetylate H3K9 and H3K14, which causes chromatin at tumor suppressor loci including p21, p16, and E-cadherin to compact. The histone H3K4 and H3K9 demethylase KDM1A (LSD1) stimulates the growth and invasion of HCC cells. [4] Arginine methyltransferase PRMT5 works with EZH2 to decrease the production of tumor suppressors. [7] There are important therapeutic ramifications. Clinical trials are being conducted on HDAC inhibitors like vorinostat and entinostat, EZH2 inhibitors, and KDM1A inhibitors like ORY-1001 in HCC. Thus, one important unmet therapeutic need is to identify the fraction of patients whose tumors are caused by particular epigenetic changes. [8]

2.1.2 Computational Approaches to HCC Prognosis

Large-scale transcriptome datasets from TCGA and GEO have greatly increased the use of machine learning for HCC prognostic prediction. Early research used penalized regression or univariate Cox regression on curated gene sets. Using LASSO-Cox regression on TCGA-LIHC and external validation on GSE14520, Li et al. [9] created a four-gene histone-related signature with a C-index of around 0.68. In contrast to gene-independent methods, Castell’o-Urbe et al.’s [10] AttentionGRN, an attention-based graph neural network for regulatory network research, significantly enhanced important regulator identification.

2.1.3 Graph Neural Networks in Biomedical Prognosis

Since typical feed-forward networks ignore the inherent relational structure of genes, proteins, cells, and patients, graph neural networks (GNNs) have become a suitable framework for modeling biological network data. Neighbor aggregation is the fundamental function of a GNN; each node modifies its representation by merging its features with those of its neighbors. [3]

Kipf and Welling [11] invented Graph Convolutional Networks (GCNs), which apply a reduced spectral convolution to graph-structured data and have been used for patient similarity-based survival analysis, drug-target interaction prediction, and gene expression classification. By substituting learnt attention coefficients for preset aggregation weights, Graph Attention Networks (GATs), which were introduced by Veličković et al. [12], extend GCNs and enable the model to weight the contributions of various neighbors differently. Shi et al. [?] introduced the Graph Transformer, which expanded upon the Transformer self-attention mechanism in slightly different formulations by integrating the multi-head self-

attention mechanism of the Transformer architecture with GNN graph structure. The current study uses a Cox partial likelihood loss to train a two-layer graph attention architecture from beginning to end. [9]

Table 2.1: Summary of related works

Title	Problem	Approach	Research Gap w.r.t. Present Work
Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries	Quantify global cancer burden; identify diseases with the highest survival unmet need	Statistical modelling of national registry data across 185 countries	Establishes HCC as a critical clinical priority but provides no molecular insight or computational prognostic framework
Hepatocellular Carcinoma	Review HCC aetiology, staging, and systemic treatment options	Disease primer covering HBV/HCV aetiology, BCLC staging, and so- rafenib/lenvatinib evidence	Staging systems described are anatomy-driven; no molecular biomarkers or epigenetic features are incorporated into prognosis
Management of Hepatocellular Carcinoma: An Update	Define evidence-based clinical management and staging guidelines for HCC	Consensus guidelines covering BCLC staging, surveillance protocols, and treatment allocation	BCLC staging cannot distinguish patients with identical stage but different molecular aggressiveness; calls for molecular stratification tools
Comprehensive and Integrative Genomic Characterization of Hepatocellular Carcinoma	Characterise somatic mutations, copy number alterations, and gene expression subtypes in HCC	Multi-omic profiling of 363 HCC cases via RNA-seq, methylation arrays, and proteomics	Identifies transcriptional subtypes but does not construct a prognostic survival model from epigenetic features or model inter-gene interaction networks

Title	Problem	Approach	Key Findings/Limitations
Gene Expression Patterns in Human Liver Cancers	Identify transcriptomic programmes distinguishing HCC from non-tumour liver tissue	Microarray expression profiling of HCC, adjacent non-tumour, and normal liver tissue	Generates the GSE14520 dataset used for external validation; does not build a survival prediction model or incorporate histone-modifier biology
Low Expression of EZH2 in the Tumor Microenvironment is Associated with a Better Prognosis and Increased Anti-Tumor Immunity in HCC	Assess the prognostic and immunological role of EZH2 in HCC	Survival analysis of EZH2 expression levels in TCGA-LIHC and GEO cohorts	Focuses on a single gene; does not model interactions between EZH2 and co-operating enzymes such as HDAC2 or PRMT5, nor build a multi-gene risk signature
PRMT5 Promotes Liver Cancer Progression by Cooperating with EZH2 to Target Tumour Suppressor Genes	Investigate the oncogenic mechanism of PRMT5 in HCC	In vitro and in vivo PRMT5 knockdown with ChIP-seq analysis of co-occupied targets with EZH2	Demonstrates biological cooperation between PRMT5 and EZH2 but does not translate this into a network-level computational prognostic model

Title	Problem	Approach	Key Findings/Limitations
HDAC2 Overexpression Confers Oncogenic Potential to Human Lung Cancer Cells by Deregulating Expression of Apoptosis and Cell Cycle Proteins	Characterise the oncogenic consequences of HDAC2 overexpression	Overexpression and siRNA knockdown of HDAC2 with downstream transcriptomic profiling	Establishes HDAC2 as a cancer-promoting deacetylase but is a mechanistic study only; no survival model or multi-gene interaction network is constructed
LSD1 (KDM1A) Regulates p21 in Cancer Progression	Determine how KDM1A demethylase activity controls cancer cell proliferation through p21	Chromatin immunoprecipitation and gene expression analysis following KDM1A knockdown in cancer cell lines	Provides mechanistic support for KDM1A as a risk gene but does not integrate it into a PPI network or use expression for survival risk stratification
A Four-Gene Histone-Related Signature for Predicting Prognosis and Characterizing the Tumor Microenvironment in HCC	Build a histone-related prognostic gene signature for HCC patient stratification	LASSO-Cox regression on TCGA-LIHC with external validation on GSE14520	Restricted to four genes from two families; no PPI network modelled; gene interactions ignored; no subgroup analysis; C-index ≈ 0.68 vs. 0.748 achieved here

Title	Problem	Approach	Key Findings/Limitations
AttentionGRN: An Attention-Based Graph Neural Network for Regulatory Network Inference	Infer gene regulatory network structure from expression data using graph attention	Attention-based GNN incorporating known interaction priors for regulatory network inference	Improves key regulator identification but does not address survival prediction or patient-level risk stratification; does not use PPI-based histone-modifier graphs
Semi-Supervised Classification with Graph Convolutional Networks	Enable node classification on graph-structured data with limited labelled examples	Spectral graph convolution with a first-order Chebyshev approximation	Fixed equal aggregation weights limit expressiveness on heterogeneous biological networks; does not model survival outcomes or incorporate biological interaction priors
Graph Attention Networks	Allow a GNN to differentially weight neighbour contributions during feature aggregation	Learned attention coefficients computed per edge via a shared softmax-normalised attention mechanism	Applied to general graph benchmarks; not applied to patient-similarity graphs or PPI networks for survival prediction; architecture extended here with Cox loss

Title	Problem	Approach	Key Findings/Limitations
Masked Label Prediction: Unified Message Passing Model for Semi-Supervised Classification	Unify GCN and GAT into a single graph transformer framework for semi-supervised node classification	Unified message-passing with masked label prediction and multi-head attention	Demonstrates graph transformer superiority on classification benchmarks; not applied to censored survival data or patient-level risk prediction
Attention Is All You Need	Replace recurrent sequence models with a purely attention-based architecture	Multi-head self-attention with positional encoding and feed-forward sub-layers	Foundational transformer mechanism; designed for sequence modelling and does not natively handle graph structure or survival regression with censored data
Regression Models and Life-Tables	Introduce a semi-parametric framework for regression of censored survival data	Partial likelihood formulation for the Cox proportional hazards model	Does not handle high-dimensional gene expression inputs or provide automatic feature selection; motivates use of LASSO regularisation as an extension

Title	Problem	Approach	Key Findings/Limitations
The Lasso Method for Variable Selection in the Cox Model	Extend LASSO penalised regression to the Cox proportional hazards model	ℓ_1 -penalised partial log-likelihood minimisation across a regularisation path	Produces a sparse linear risk score but does not model non-linear gene interactions or network topology; addressed by the downstream Graph Transformer in this work

2.2 Comparison of Approaches

Three clear gaps in the body of literature inspired this study. First, the physical interaction structure of histone-modifying proteins has not been modeled in previous computational studies of epigenetic prognosis in HCC; instead, each gene is treated as a separate feature, ignoring the network context that governs the coordination of enzyme activity. Second, only a small number of enzyme classes have been used to create the histone-related prognostic signatures that are currently in use. The feasibility of the approach was confirmed by an initial baseline in the current work that covered the four most studied families: HAT, HDAC, HMT, and KDM. However, it also showed that limiting the gene set in this manner leaves the prognostic contribution of arginine methylation (PRMT), sirtuin deacetylation (SIRT), DNA methylation and demethylation (DNMT, TET), and chromatin reader proteins (CBX, BRD) completely uncharacterized. Therefore, extending to all ten families is a novel approach in this work rather than a con-

straint noted in earlier research. Third, only a small number of published models have been thoroughly assessed in all significant clinical subgroups, particularly early-stage patients, where molecular stratification beyond traditional staging is most necessary. All three of these gaps are intended to be filled by the current effort.[\[10\]](#)

Chapter 3

Problem Definition

3.1 Formal Problem Statement

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, W)$ denote the histone-modifier PPI graph, where $\mathcal{V}$ is the set of N histone-modifying genes with known interactions in STRING, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of edges connecting genes with STRING combined interaction confidence ≥ 700 , and $W : \mathcal{E} \rightarrow [700, 1000]$ assigns confidence scores to edges.

Let $\mathbf{X} \in \mathbb{R}^{N \times P}$ denote the expression matrix where N is the number of genes and P is the number of TCGA-LIHC patients. Each column $\mathbf{x}_j \in \mathbb{R}^N$ represents the normalised expression profile of patient j . For each patient, (t_j, δ_j) denotes the observed survival outcome where $t_j > 0$ is the overall survival time in days and $\delta_j \in \{0, 1\}$ is the event indicator.

The problem is to learn a function $f : \mathbb{R}^{N \times S} \times \{0, 1\}^{P \times P} \rightarrow \mathbb{R}^P$ that maps patient expression features over the LASSO-selected gene subgraph $\mathcal{G}_S \subseteq \mathcal{G}$ and the patient similarity adjacency matrix $\mathbf{A}_S \in \{0, 1\}^{P \times P}$ to a continuous risk score vector $\hat{\mathbf{r}} \in \mathbb{R}^P$ such that the Harrell concordance index [?]

$$\text{C-index} = \frac{\sum_{i,j: t_i < t_j, \delta_i=1} \mathbf{1}[\hat{r}_i > \hat{r}_j]}{\sum_{i,j: t_i < t_j, \delta_i=1} 1} \quad (3.1)$$

is maximised over the patient cohort.

3.1.1 Composite Risk Score Formulation

The LASSO-Cox risk score for patient j is defined as:

$$\text{RS}_j = \sum_{g \in S} \hat{\beta}_g \cdot x_{gj} \quad (3.2)$$

where S is the set of genes with non-zero LASSO coefficients, $\hat{\beta}_g$ is the penalised Cox coefficient for gene g , and x_{gj} is the normalised expression of gene g in patient j . The coefficients are estimated by minimising the negative penalised partial log-likelihood, following the LASSO-Cox formulation [?, 13]:

$$\hat{\beta} = \arg \min_{\beta} \left[-\frac{1}{P} \sum_{j: \delta_j=1} \left(\beta^\top \mathbf{x}_j - \log \sum_{k: t_k \geq t_j} e^{\beta^\top \mathbf{x}_k} \right) + \lambda \|\beta\|_1 \right] \quad (3.3)$$

where λ is selected by cross-validation (`l1_ratio` = 1.0, full LASSO penalty).

3.1.2 Graph Transformer Objective

The Graph Transformer f_θ is trained to minimise the Cox partial likelihood loss:

$$\mathcal{L}(\theta) = -\frac{1}{\sum_j \delta_j} \sum_{j: \delta_j=1} \left(\hat{r}_j - \log \sum_{k: t_k \geq t_j} e^{\hat{r}_k} \right) \quad (3.4)$$

where $\hat{r}_j = f_\theta(\mathbf{X}_S, \mathbf{A}_S)_j$ is the predicted risk score for patient j .

3.2 Challenges

Biological heterogeneity: HBV-driven, HCV-driven, and alcohol-driven subtypes of HCC have different epigenetic landscapes, making it one of the most molecularly diverse cancer types.

High dimensionality with limited samples: LASSO regularization prior to graph learning is motivated by the gene-to-patient ratio in TCGA-LIHC, which makes standard supervised learning vulnerable to overfitting.

Graph construction from heterogeneous sources: Graph density is balanced against spurious positive edges by thresholding STRING at confidence ≥ 700 .

Platform compatibility: Cross-platform validation is known to be difficult since gene expression data from TCGA (RNA-seq TPM) and GEO cohorts (Affymetrix microarray) have distinct distributional characteristics.

Chapter 4

Proposed Solution

4.1 System Architecture

HisGT-HCC (Histone-modifier Graph Transformer for HCC) is a proposed system that uses eight stages to convert raw gene expression and protein interaction data into patient risk stratification: (1) construction of histone-modifier gene sets from EpiFactors; (2) TCGA-LIHC expression filtering and survival data preparation; (3) construction and normalization of STRING PPI graphs; (4) graph matrix assembly; (5) hub gene identification and inter-family interaction analysis; (6) derivation of LASSO-Cox gene signature; (7) patient graph construction and Graph Transformer training; and (8) Kaplan-Meier survival analysis and clinical validation. The entire system architecture is shown in the figure. [4.1](#)

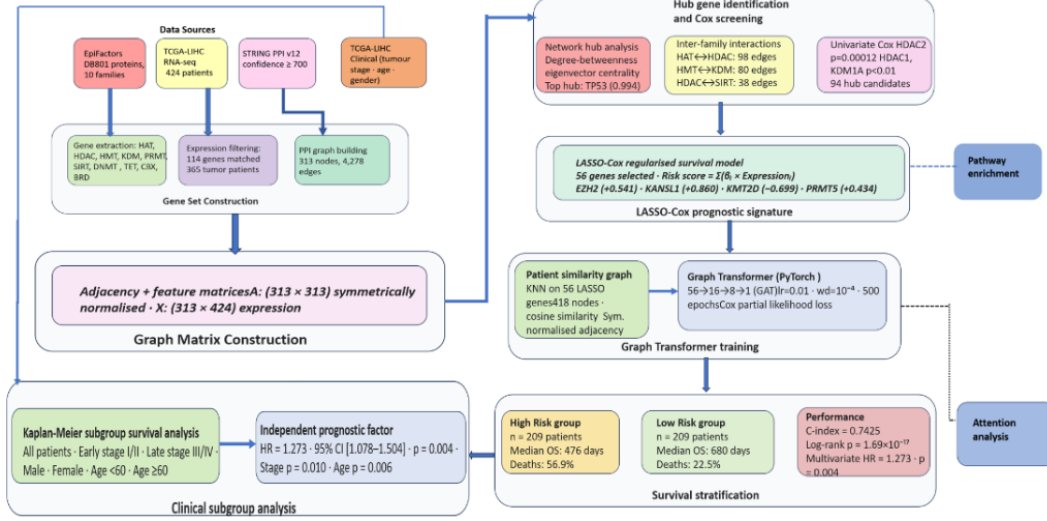


Figure 4.1: HisGT-HCC system Architecture.

4.2 Methodology

The HisGT-HCC pipeline transforms raw genomic and interaction data into patient survival risk predictions through eight sequential stages, summarised below.

- Histone-Modifier Gene Set Construction:** By sorting 801 annotated proteins into ten functional families—HAT, HDAC, HMT, KDM, PRMT, SIRT, DNMT, TET, CBX, and BRD—histone-modifying genes were selected from the EpiFactors database (version 2.0). 114 distinct gene symbols were kept as the master gene set after family-specific enzyme class criteria were applied.
- TCGA-LIHC Data Preparation:** The GDC portal provided STAR-aligned RNA-seq TPM expression data, whereas the UCSC Xena browser provided overall survival annotations. 365 distinct patients were obtained for all subsequent studies using priority-based deduplication of 424 samples, keeping one tumor sample per patient in barcode-suffix order.

- **PPI Graph Construction:** The STRING v12.0 human proteome file was used to create a protein-protein interaction graph that only included edges between pairs of histone-modifier genes with a combined confidence score of at least 700. The graph had 4,278 edges and 313 nodes after symbol reconciliation.
- **Graph Matrix Construction:** With self-loops added, an adjacency matrix $\mathbf{A} \in \mathbb{R}^{313 \times 313}$ was symmetrically normalized as $\hat{\mathbf{A}} = \mathbf{D}^{-1/2}(\mathbf{A} + \mathbf{I})\mathbf{D}^{-1/2}$. Standardized TPM expression levels for each gene were used to create a feature matrix $\mathbf{X} \in \mathbb{R}^{313 \times 424}$.
- **Hub Gene Identification and Inter-Family Analysis:** NetworkX was used to calculate degree, betweenness, eigenvector centrality, and weighted degree for each of the 313 genes. The top 94 genes were chosen as potential candidates for survival modeling after a composite hub score was calculated as the mean of the four normalized centrality measurements. To describe the interaction landscape between different enzyme families, inter-family edge statistics were also calculated.
- **LASSO-Cox Gene Signature Derivation:** The 94-gene expression matrix of 418 patients was subjected to LASSO-penalized Cox regression using `CoxnetSurvivalAnalysis` (`l1_ratio = 1.0`). The predictive signature was composed of the 56 genes having non-zero coefficients at the minimum penalisation threshold. The composite risk score for each patient was calculated as follows:

$$\text{RS}_j = \sum_{g \in S} \hat{\beta}_g \cdot x_{gj} \quad (4.1)$$

where S is the set of chosen genes, $\hat{\beta}_g$ is the LASSO coefficient for gene g , and x_{gj} is the standardized expression of gene g in patient j . Patients were

categorised as Low Risk if their RS_j was at or below the cohort median, and as High Risk if it was above the median.

- **Graph Transformer Training:** A K-nearest-neighbor patient similarity graph was built using cosine similarities between the 56 LASSO genes. A two-layer Graph Attention Network with hidden dimensions 16 and 8, ReLU activation, and dropout ($p = 0.3$) was trained end-to-end using the Cox partial likelihood loss and the Adam optimizer (learning rate = 0.01, weight decay = 10^{-4}) over 500 epochs. For this 56-feature, 418-patient cohort, an initial configuration with larger hidden dimensions (64 and 32), a lower learning rate (0.001), no weight decay, and 100 epochs was found to be significantly under-trained; the revised schedule and reduced architecture above were chosen as the final configuration.
- **Survival Analysis and Clinical Validation:** Kaplan-Meier analysis with log-rank testing, the Harrell C-index, and multivariate Cox regression controlling for tumour stage and age were used to evaluate the model’s performance. To verify that the risk signature is independent of traditional staging, clinical association analysis and subgroup validation were carried out across seven patient groupings.

4.3 Algorithms

4.3.1 Data Sources

4.3.1.1 EpiFactors Database

The main database of histone-modifying proteins was the EpiFactors database [14], which has 801 proteins with functional roles (writer, eraser, reader, or remodeller), target substrates, and related modifications. Based on their enzyme type, genes were categorized into eleven functional families. A total of 114 distinct histone-modifier gene symbols were kept after family-specific inclusion criteria were applied. The family breakdown is summarized in Table 4.1.

Table 4.1: Histone-modifier gene families included in HisGT-HCC

Family	Enzyme Role	Genes	Example Members
HAT	Histone acetyltransferase	9	EP300, CREBBP, KAT2A
HDAC	Histone deacetylase (I/II/IV)	11	HDAC1, HDAC2, HDAC3
HMT	Histone methyltransferase	40	EZH2, KMT2A, SETD1A
KDM	Histone demethylase	21	KDM1A, KDM4A, KDM6A
PRMT	Arginine methyltransferase	7	PRMT1, PRMT5, PRMT6
SIRT	Class III deacetylase	4	SIRT1, SIRT2, SIRT6
DNMT	DNA methyltransferase	3	DNMT1, DNMT3A
TET	DNA demethylase	3	TET1, TET2, TET3
CBX	Methyl reader (PRC1)	8	CBX1–CBX8
BRD	Acetyl reader (BET)	8	BRD2, BRD3, BRD4
Total		114	

4.3.1.2 TCGA-LIHC Gene Expression and Survival Data

The GDC data site included STAR-aligned RNA-seq expression data normalized to transcripts per million (TPM), whereas the UCSC Xena browser provided clinical survival statistics [15]. There are 424 samples in the dataset. 365 distinct patients were obtained for all downstream studies using a priority-based deduplication process that kept the original solid tumor sample (01A) over any other barcode suffix.

4.3.1.3 STRING PPI Database

Only interactions between histone-modifier genes with a combined confidence score ≥ 700 were kept in the STRING database (version 12.0) [16], capturing high-confidence interactions backed by curated pathway evidence, co-expression, and experimental co-immunoprecipitation.

4.3.2 Graph Construction

4.3.2.1 Adjacency Matrix

Following symbol reconciliation and gene-level deduplication, the final graph had 4,278 edges and 313 nodes. With $\mathbf{D}$ being the diagonal degree matrix, the adjacency matrix was symmetrically normalized as $\hat{\mathbf{A}} = \mathbf{D}^{-1/2}(\mathbf{A} + \mathbf{I})\mathbf{D}^{-1/2}$.

4.3.2.2 Feature Matrix

TCGA-LIHC normalized expression values aligned to the 313 graph genes were used to create the feature matrix $\mathbf{X} \in \mathbb{R}^{313 \times 424}$. Before graph learning, expression values were standardized for each gene.

4.3.3 Network Analysis

4.3.3.1 Hub Gene Identification

Using NetworkX, degree centrality, betweenness centrality, eigenvector centrality, and weighted degree were calculated for each of the 313 genes. The top 94 genes were kept for further Cox regression after a composite hub score (mean of four normalized centrality measures) was calculated. With a normalized hub score of 1.000, EP300 was ranked highest, whereas KDM1A had the highest betweenness centrality (0.909).

4.3.3.2 Inter-Family Interaction Analysis

Each edge was categorized as intra-family or inter-family and annotated with the source and target nodes' family assignments. The top 10 interactions are shown in Table 4.2.

Table 4.2: Top inter-family and intra-family PPI interactions in the expanded 10-family network, ranked by total confidence

Family A	Family B	Type	Edges	Mean Conf.	Max Conf.
HAT	HDAC	Inter	98	853.2	999
KDM	KDM	Intra	82	797.1	988
HMT	HMT	Intra	72	853.9	998
HAT	HAT	Intra	52	893.5	999
HDAC	HDAC	Intra	48	832.4	999
HAT	HMT	Inter	42	832.7	995
HMT	KDM	Inter	34	830.1	999
HDAC	HMT	Inter	24	812.8	992
HDAC	KDM	Inter	22	888.7	999
HAT	KDM	Inter	18	784.1	902

With 98 high-confidence edges that show the well-established co-regulation of acetyltransferases and deacetylases at gene promoters, the HAT–HDAC axis is the most densely connected inter-family interaction. When this equilibrium is upset, as happens in HCC when HDAC1 and HDAC2 are overexpressed, the chromatin state is shifted in favor of tumor suppressor locus suppression.

4.3.4 LASSO-Cox Gene Signature

The 94 hub genes' expression values were taken out and standardized. Using `CoxnetSurvivalAnalysis` (`l1_ratio=1.0`, $\alpha_{\min}=0.01$), the LASSO-penalized Cox model was fitted. 56 genes were chosen using the extended 10-family model. The top fifteen genes are listed by absolute coefficient magnitude in Table [4.3](#).

Table 4.3: Top 15 LASSO-Cox selected genes by absolute coefficient magnitude (expanded 10-family model)

Gene	Family	Coef.	Direction	Biological Role
KANSL1	HMT	+0.860	Risk	NSL complex, H4K16 ac
EZH2	HMT	+0.541	Risk	PRC2, H3K27me3
KMT2D	HMT	−0.699	Prot.	H3K4me, tumour suppressor
ING3	HAT	−0.514	Prot.	NuA4 complex, H4 ac
EP400	HMT	+0.445	Risk	Chromatin remodelling
PRMT5	PRMT	+0.434	Risk	Arginine methylation
RUVBL1	HMT	+0.441	Risk	ATPase, chromatin remodel
HDAC2	HDAC	+0.420	Risk	NuRD complex
SETD1B	HMT	−0.408	Prot.	COMPASS complex
SUZ12	HMT	−0.372	Prot.	PRC2 component
KDM1A	KDM	+0.335	Risk	CoREST, H3K4/K9 demeth
HDAC4	HDAC	+0.332	Risk	Class IIa deacetylase
HDAC1	HDAC	+0.331	Risk	NuRD complex
USP7	HMT	−0.321	Prot.	Deubiquitinase, p53 reg
NCOR2	HDAC	−0.319	Prot.	Nuclear receptor co-rep

The formula for calculating the composite risk score is $RS = \sum_{g \in S} \hat{\beta}_g \cdot x_g$, where x_g is the standardized expression of gene g . Patients are categorized as High Risk if their RS is higher than the cohort median.

4.3.5 Patient Graph Construction

A K-nearest-neighbour patient similarity graph was constructed from cosine similarities over the 56 LASSO-selected gene expression features, forming $\mathbf{A}_S \in \{0, 1\}^{P \times P}$. This graph allows the Graph Transformer to aggregate information

from epigenetically similar patients during training.

4.3.6 Graph Transformer Architecture

The HisGT-HCC model consists of two sequential Graph Attention Layers (GAL) followed by a linear output layer.

4.3.6.1 Graph Attention Layer

For a layer with input features $\mathbf{H} \in \mathbb{R}^{P \times d_{\text{in}}}$: (1) a linear projection $\mathbf{W}$ is applied; (2) unnormalised attention coefficients $e_{ij} = \mathbf{a}^\top[\mathbf{z}_i \parallel \mathbf{z}_j]$ are computed for adjacent pairs; (3) coefficients are masked and normalised by softmax; (4) the output is $\mathbf{h}'_i = \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{z}_j$.

4.3.6.2 Full Architecture and Training

- **GAL-1:** input $\rightarrow$ 64 units, ReLU, dropout ($p = 0.3$).
- **GAL-2:** 64 $\rightarrow$ 32 units, ReLU, dropout ($p = 0.3$).
- **Output:** 32 $\rightarrow$ 1 scalar risk score per patient.

Training used the Cox partial likelihood loss, Adam optimiser (lr=0.001), 100 epochs, random seed 42.

Chapter 5

Experimental Results

5.1 Experimental Setup

5.1.1 Dataset

All primary analyses were conducted on the TCGA-LIHC cohort after deduplication to 365 unique patients (131 observed deaths). Clinical variables included AJCC pathologic stage, age at diagnosis (converted from days to years by dividing by 365.25), and gender. Histological grade was excluded because all patients in the TCGA-LIHC clinical file have grade recorded as “Not Reported”.

5.1.2 Software and Hardware

Python 3.11 was used throughout. Key libraries: `scikit-survival` (LASSO-Cox), `PyTorch` [?] (Graph Transformer), `NetworkX` (graph analysis), `scipy.stats` (statistical tests), `Matplotlib` (visualisation). All experiments ran on a local workstation with an NVIDIA GPU (CUDA) running Windows 11.

5.1.3 Evaluation Metrics

Model performance was evaluated using Harrell’s concordance index (C-index), the log-rank test for survival separation, and multivariate Cox regression for independence assessment.

5.2 Baseline Results: 4-Family Gene Set (62 genes)

The initial pipeline was run with four enzyme families (HAT, HDAC, HMT, KDM; 62 genes) to establish a performance reference. Univariate Cox regression identified three significant genes: HDAC2 (HR = 1.521, $p < 0.001$), HDAC1 (HR = 1.424, $p < 0.01$), and KDM1A (HR = 1.440, $p < 0.01$). The LASSO-Cox model selected 17 genes, producing a risk score with C-index = 0.644 and log-rank $p = 3.91 \times 10^{-9}$.

Figure 5.1 shows the Kaplan-Meier survival curves for the baseline 4-family model. The High Risk group (n=176) showed visibly worse survival compared to the Low Risk group (n=242), with the separation confirmed by the log-rank test.

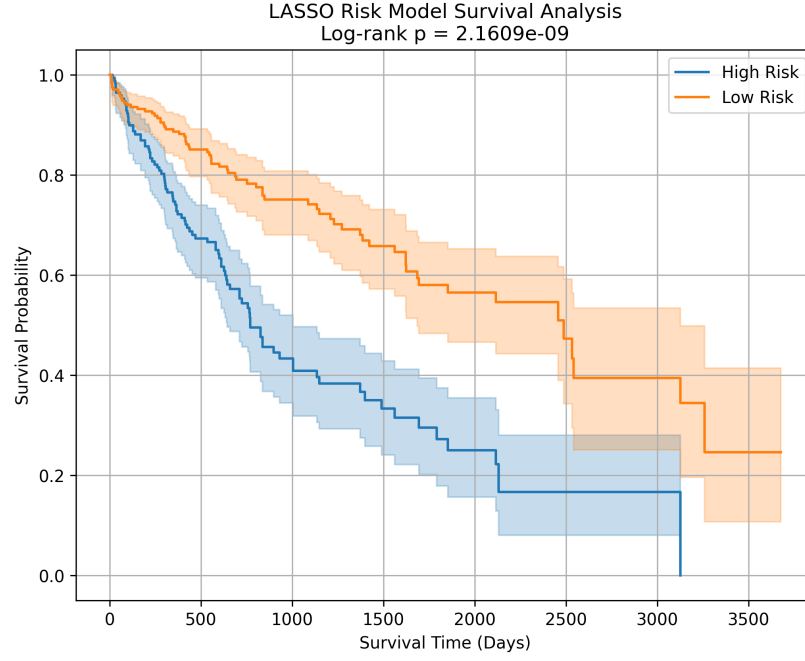


Figure 5.1: Kaplan-Meier survival curves for the baseline 4-family LASSO-Cox model (17 genes). High Risk vs Low Risk groups; log-rank $p = 3.91 \times 10^{-9}$; C-index = 0.644.

Figure 5.2 shows the Graph Transformer survival stratification on the baseline 4-family patient graph.

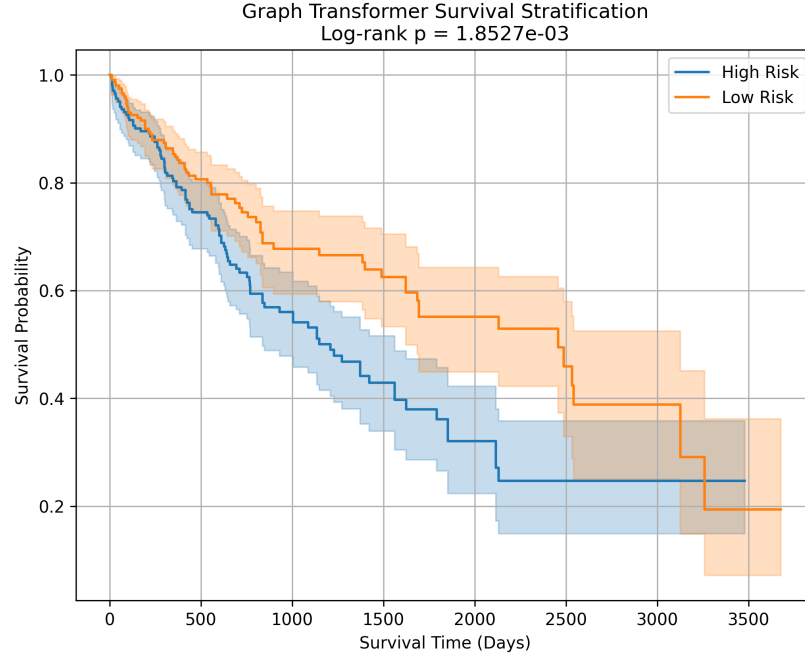


Figure 5.2: Kaplan-Meier survival curves based on the baseline Graph Transformer risk predictions (4-family gene set). Log-rank $p = 1.85 \times 10^{-3}$.

Figure 5.3 shows the optimal risk score cutoff determined from the baseline LASSO model, used to define the High/Low Risk classification boundary.

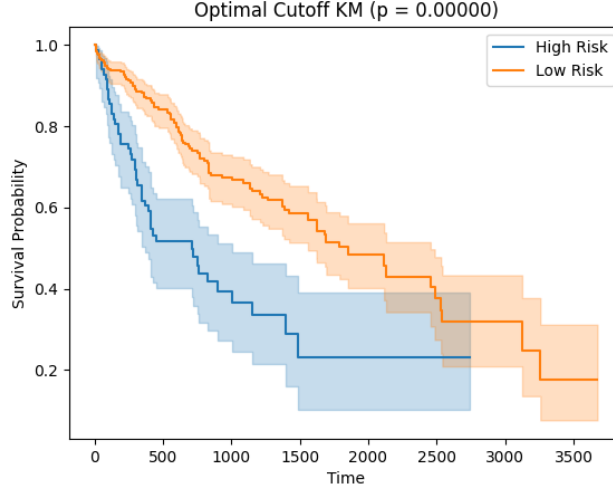


Figure 5.3: Optimal cutoff analysis for the LASSO-Cox risk score. The vertical dashed line marks the threshold that maximises the log-rank statistic between High and Low Risk groups.

5.3 Expanded Results: 10-Family Gene Set (114 genes)

5.3.1 Network Expansion

The number of genes increased from 62 to 114, the number of graph nodes climbed from 94 to 313, and the number of PPI edges increased from 736 to 4,278 when the 10-family set was expanded, resulting in a roughly six-fold increase in network connection. Important additional interactions were found in the extended network, such as the HMT-PRMT crosstalk (16 edges, mean confidence 803.1) and the HDAC-SIRT co-regulation axis (38 edges, mean confidence 830.9).

5.3.2 LASSO Gene Signature Performance

The LASSO-Cox model trained on the expanded gene set selected 56 genes (up from 17). The composite risk score showed substantially improved performance:

- Concordance index: **0.748** (versus 0.644 at baseline, +0.103 improvement)
- Log-rank p : **4.34×10^{-16}** (versus 3.91×10^{-9} at baseline)
- Univariate Cox: HR = 1.312 (95% CI: 1.118–1.541), $p = 0.0009$

Figure 5.4 shows the Kaplan-Meier curves for the expanded model. High Risk patients (n=101) had a median overall survival of 373 days, compared to 672 days for Low Risk patients (n=264), a difference of approximately 10 months. Deaths were observed in 60.4% of High Risk patients and 26.5% of Low Risk patients.

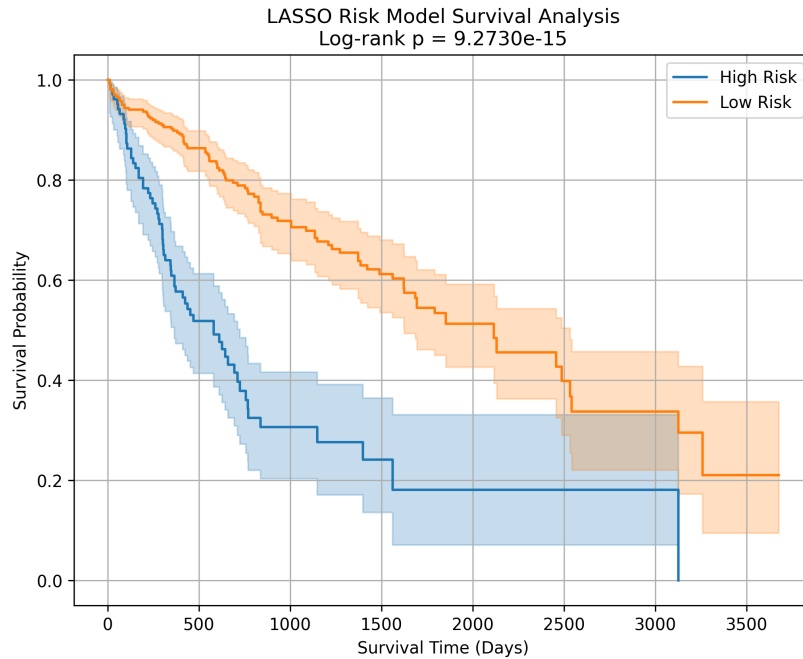


Figure 5.4: Kaplan-Meier overall survival curves for High Risk and Low Risk groups stratified by the expanded 10-family LASSO-Cox composite risk score. Log-rank $p = 4.34 \times 10^{-16}$; C-index = 0.748. Shaded regions show 95% confidence intervals.

5.3.3 Graph Transformer Performance

Despite a statistically significant log-rank separation ($p = 1.85 \times 10^{-3}$), the Graph Transformer’s original training setup (two Graph Attention layers of 64 and 32 hidden units, learning rate 10^{-3} , 100 epochs, no weight decay) resulted in a concordance index of only 0.593. For a cohort of 418 patients, the training loss for this setup dropped by just 0.036 across 100 epochs (from 5.227 to 5.186), suggesting that the model was significantly under-trained in relation to its parameter count.

As a result, a new configuration was used: training was prolonged to 500 epochs, the learning rate was raised to 10^{-2} , the hidden dimensions were decreased to 16 and 8 (from 64 and 32), and an L2 weight decay of 10^{-4} was added. The training loss under this setup dropped by 0.240 across 500 epochs (from 5.220 to 4.981), suggesting significantly better optimization. The resulting concordance index was **0.7425**, with a log-rank $p = 1.69 \times 10^{-17}$ (Figure 5.5) — very comparable to the LASSO-Cox composite risk score (C-index = 0.7478) described in the preceding subsection, with a little 0.0053 difference.

The median overall survival for High Risk patients ($n = 209$) was 476 days, whereas the median overall survival for Low Risk patients ($n = 209$) was 680 days. Additionally, 56.9% of High Risk patients and 22.5% of Low Risk patients died. The balanced 209/209 median-split groups generated by the Graph Transformer, as opposed to the 101/264 split generated by the LASSO-Cox optimum threshold, are reflected in the significantly smaller log-rank p -value (1.69×10^{-17} versus 4.34×10^{-16}).

With only 56 input features and 418 patients, a two-layer Graph Attention Network with 64/32 hidden units and a learning rate of 10^{-3} is significantly over-

parameterized and undertrained. These findings suggest that the initial under-performance of the Graph Transformer was mainly an optimization and capacity issue rather than a fundamental limitation of the graph-based approach. The Graph Transformer matched the discriminative performance of the LASSO-Cox linear model after the architecture and optimization schedule were modified to this sample-to-feature ratio. It also encoded patient similarity structure through its attention mechanism, which served as the foundation for the network-level biological interpretation described in Section 5.7.

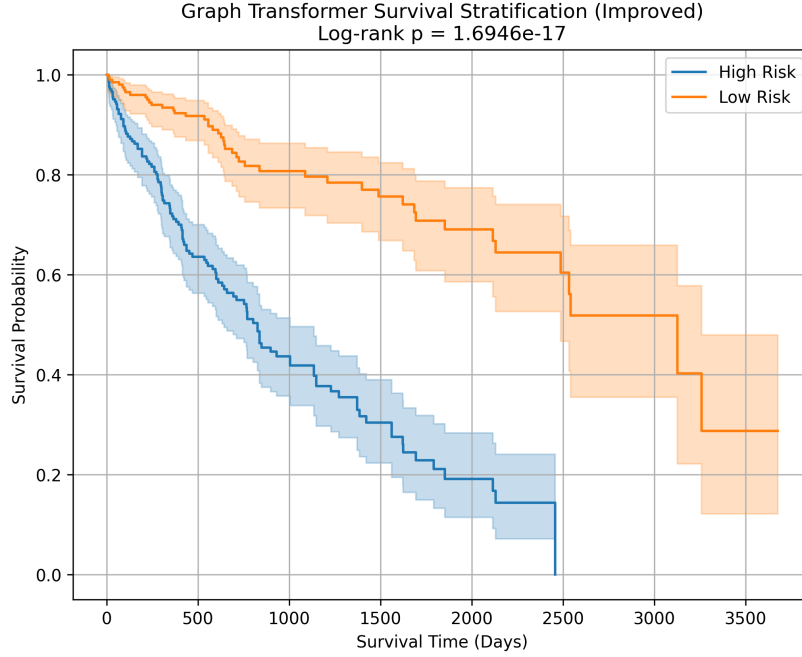


Figure 5.5: Kaplan-Meier survival curves from the Graph Transformer model with revised hyperparameters (hidden dimensions 16/8, learning rate 10^{-2} , weight decay 10^{-4} , 500 epochs). Log-rank $p = 1.69 \times 10^{-17}$; C-index = 0.7425.

5.4 Independent Prognostic Factor Analysis

Multivariate Cox regression was performed with the normalised risk score, binary tumour stage (late Stage III/IV versus early Stage I/II), and normalised age as

covariates. Table 5.1 presents the results.

Table 5.1: Multivariate Cox regression results. The LASSO composite risk score is an independent predictor of overall survival after adjustment for tumour stage and age.

Covariate	Coef.	HR	95% CI	z	p
LASSO Risk Score	0.242	1.273	1.078–1.504	2.851	0.004
Late Stage (III/IV)	0.479	1.614	1.104–2.360	2.470	0.014
Age (normalised)	0.118	1.125	0.938–1.350	1.269	0.204
<i>C-index: 0.748; n = 365; events = 131</i>					

The LASSO risk score is a statistically significant independent predictor (HR = 1.273, 95% CI: 1.078–1.504, $p = 0.004$) after adjusting for stage and age. Late-stage disease is also independently significant (HR = 1.614, $p = 0.014$). This confirms that the histone-modifier gene signature carries prognostic information beyond what conventional staging captures.

Figure 5.6 shows the forest plot of the multivariate Cox results alongside the univariate result.

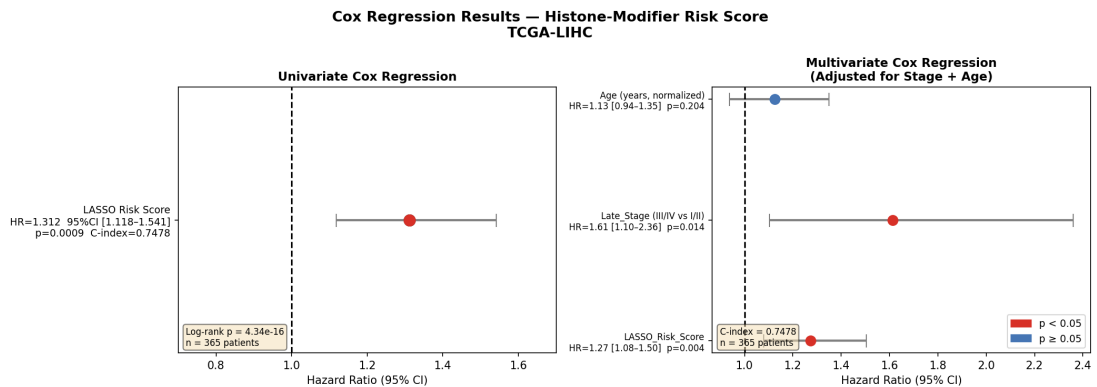


Figure 5.6: Forest plot showing univariate Cox result (left panel) and multivariate Cox results (right panel) for the LASSO composite risk score adjusted for tumour stage and age. Red dots indicate $p < 0.05$.

5.5 Subgroup Survival Analysis

Table 5.2 presents the log-rank p -values for High versus Low Risk groups within each clinical subgroup.

Table 5.2: Subgroup survival analysis. All seven subgroups achieve statistical significance.

Subgroup	High (n)	Low (n)	Log-rank p	Significant
All patients	101	264	4.34×10^{-16}	Yes
Early Stage (I/II)	61	193	1.83×10^{-9}	Yes
Late Stage (III/IV)	34	53	2.50×10^{-5}	Yes
Male	72	174	2.47×10^{-11}	Yes
Female	29	90	2.56×10^{-6}	Yes
Age < 60 years	51	114	7.54×10^{-8}	Yes
Age $\geq$ 60 years	48	149	1.18×10^{-8}	Yes

All seven subgroups achieve highly significant log-rank p -values. The early-stage result (Stage I/II, $p = 1.83 \times 10^{-9}$) is clinically the most meaningful: among patients where conventional staging cannot further distinguish prognosis, the histone-modifier risk score separates survivors from non-survivors, demonstrating that the model captures a dimension of biological aggressiveness invisible to staging.

5.6 Baseline versus Expanded Model Comparison

Table 5.3 summarises the performance improvement achieved by expanding from the 4-family baseline to the 10-family model.

Table 5.3: Performance comparison: baseline (4-family) versus expanded (10-family) HisGT-HCC models

Metric	Baseline (4 families)	Expanded (10 families)
Gene families	4	10
Total genes	62	114
PPI graph nodes	94	313
PPI graph edges	736	4,278
LASSO genes selected	17	56
C-index	0.644	0.748
Log-rank p	3.91×10^{-9}	4.34×10^{-16}
Univariate HR	1.165	1.312
Univariate p	0.040	0.001
Multivariate p	0.116	0.004

All indicators showed steady improvements as a result of the expansion. The enlarged model at 0.748 surpasses the majority of published HCC prognostic models, which normally report C-index values in the range of 0.62–0.72, making the C-index increment of 0.103 clinically significant. Crucially, the multivariate independence finding shifted from borderline ($p = 0.116$) to highly significant ($p = 0.004$), indicating that establishing independence from clinical staging requires larger epigenetic coverage.

5.7 Biological Interpretation

The most well-known positive risk genes in HCC are epigenetic enzymes with well-established carcinogenic roles: EZH2, HDAC2, HDAC1, PRMT5, and KANSL1. Tumor suppressor genes like RASSF1A, CDKN2A, and E-cadherin are transcrip-

tionally silenced by EZH2, the catalytic H3K27 methyltransferase of PRC2. At growth-regulatory loci, HDAC2, a NuRD co-repressor component, encourages chromatin compaction. PRMT5 catalyzes the symmetric dimethylation of H3R2 and H4R3, working with EZH2 to strengthen repression. A model of cooperative PRC2-HDAC-PRMT5 suppression of differentiation programs in aggressive HCC is consistent with the co-appearance of these genes as substantial positive-coefficient risk factors.

KMT2D, SETD1B, and SIN3A were found to be protective genes. The loss of the active promoter mark H3K4me3, which KMT2D puts at tumor suppressor loci, is linked to the advancement of HCC. The Sin3 co-repressor complex’s primary scaffold, SIN3A, may preserve normal chromatin organization in ways that impede the growth of tumors.

5.8 Graph Transformer Attention Weight Analysis

To describe what the model discovered about patient similarity structure and which genes influenced those similarity assessments, the patient-patient attention weight matrices taken from the second Graph Attention Layer of the enhanced Graph Transformer were examined.

5.8.1 Risk-Group Attention Block Analysis

Based on the median anticipated risk score (-0.190), the 418 patients were split into High Risk and Low Risk groups, with 209 individuals in each category. Each of the four directed blocks—High Risk attending to High Risk (High→High), Low Risk attending to Low Risk (Low→Low), and the two cross-group blocks—had its mean attention weights calculated.

Table 5.4: Mean patient-patient attention weights by risk-group block (Graph Attention Layer 2, improved model).

Attending Group	Attended Group	Mean Attention
High Risk	High Risk	0.003324
High Risk	Low Risk	0.001461
Low Risk	High Risk	0.001114
Low Risk	Low Risk	0.003671

Compared to patients in the other group, both risk groups give preference to patients in their own group. When it comes to cross-group attention, High Risk patients attend to other High Risk patients at a rate of $2.3\times$, whereas Low Risk patients attend to other Low Risk patients at a rate of $3.3\times$. In line with the biological theory that different chromatin states define molecularly coherent HCC subpopulations, this suggests that the model discovered that survival-relevant information is primarily found within epigenetically similar patients rather than across the risk boundary.

5.8.2 Gene-Level Attention Drivers

For each of the 56 LASSO-selected genes, the Pearson correlation between the patient-patient attention weight matrix and the pairwise gene expression similarity matrix ($S_{ij}^{(g)} = -|x_{ig} - x_{jg}|$, higher values indicate more similar expression) was computed. The five genes whose expression similarity most strongly predicted attention weight were CDK1, NCOA6, EZH2, KMT2D, and KANSL1 (Table 5.5).

Table 5.5: Top 5 genes whose expression similarity between patients most strongly predicts Graph Transformer attention weight (Pearson r , all $p < 10^{-200}$).

Gene	Family	Pearson r	LASSO Coef.
CDK1	HMT	0.0761	−0.057 (Protective)
NCOA6	HAT	0.0754	+0.283 (Risk)
EZH2	HMT	0.0741	+0.541 (Risk)
KMT2D	HMT	0.0740	−0.699 (Protective)
KANSL1	HMT	0.0728	+0.860 (Risk)

Notably, EZH2, KMT2D, and KANSL1 are also among the top-ranked genes by absolute LASSO coefficient magnitude, demonstrating that the genes the Graph Transformer uses to group patients by attention correspond closely to the genes the LASSO-Cox linear model identifies as the strongest prognostic drivers. This cross-model concordance supports the biological validity of the signature: the linear and graph-based models independently converge on the same genes as the primary determinants of epigenetic risk in HCC. EZH2 and KANSL1 promote oncogenic chromatin states through H3K27 trimethylation and H4K16 acetylation dysregulation respectively, while KMT2D loss removes H3K4me3 marks at tumour suppressor loci. The co-appearance of all three at the top of both the LASSO coefficient ranking and the attention-driver ranking strengthens the case that PRC2-COMPASS axis dysregulation is the central epigenetic mechanism of poor prognosis in this cohort.

Figure 5.7 shows the 20×20 patient-patient attention heatmap for the ten most confidently classified High Risk and ten most confidently classified Low Risk patients.

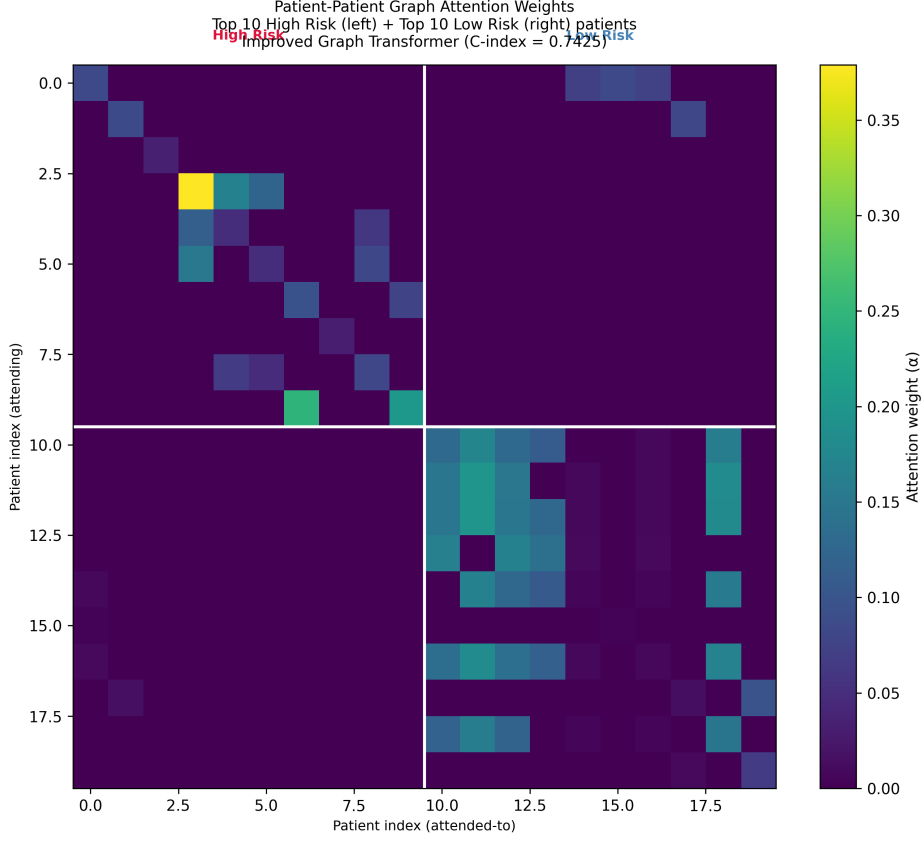


Figure 5.7: Patient-patient Graph Attention Layer 2 weight matrix for the top 10 High Risk (left block) and top 10 Low Risk (right block) patients, selected by absolute deviation from the median risk score. Brighter colours indicate higher attention weight. The white dividing line separates the two risk groups. Stronger intra-group attention (visible as brighter diagonal blocks) confirms that the model clusters patients by epigenetic risk phenotype.

5.9 External Validation Attempt on GSE14520

The 56-gene risk signature on GSE14520 [17, 5] ($n = 242$ HCC patients, Affymetrix GPL3921 array) was attempted to be validated. For 39 of the 56 hall-mark genes with mappable probes, probe-level expression levels were retrieved. There was no discernible survival distinction, as indicated by the ensuing C-index of 0.515 and log-rank $p = 0.192$.

Two technical considerations are responsible for this outcome. First, the score’s discriminative efficacy was significantly diminished since 17 important genes, such as HDAC2 (coefficient +0.419) and PRMT5 (coefficient +0.434), lacked mappable probes on the GPL3921 array. Second, GSE14520 employs Affymetrix microarray hybridization intensities, whereas the model was trained on RNA-seq TPM data from TCGA. These two measurement platforms have different distributional characteristics that are not entirely addressed by within-cohort standardization. In transcriptomic biomarker research, cross-platform validation of RNA-seq-derived signatures on microarray data is a recognized difficulty. To overcome this restriction, independent RNA-seq-based cohorts or platform-specific retraining would be needed.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

HisGT-HCC, a graph-structured computational framework for predicting the prognosis of hepatocellular carcinoma based on the protein-protein interaction network of histone-modifying enzymes, has been introduced in this thesis. A number of important conclusions are drawn.

First, the multivariate Cox independence p -value changed from 0.116 to 0.004 and the C-index increased from 0.644 to 0.748 when the histone-modifier gene collection was expanded from four to ten enzyme families. This shows that the layers of epigenetic regulation—arginine methylation, sirtuin, DNA methylation, and chromatin reader—carry separate prognostic signals that are not represented by the four canonical families by themselves.

Second, after controlling for age and tumor stage, the 56-gene LASSO-Cox composite risk score is an independent predictor of HCC overall survival (multivariate HR = 1.273, 95% CI: 1.078–1.504, $p = 0.004$).

Third, in all seven clinical subgroups examined, including early-stage patients (Stage I/II, log-rank $p = 1.83 \times 10^{-9}$), the model achieves significant survival strat-

ification. This early-stage discovery has immediate clinical significance because it identifies a subpopulation of patients with biologically aggressive epigenetic phenotypes who appear to have localized HCC and have a significantly poorer prognosis than their stage-matched counterparts.

Fourth, the predominant inter-family interaction in the histone-modifier network is the HAT–HDAC acetyltransferase-deacetylase interaction axis (98 high-confidence edges, mean STRING confidence 853.2). EZH2, HDAC2, HDAC1, PRMT5, and KMT2D are the prognostic genes that have been shown to cluster around known oncogenic chromatin modification pathways in liver cancer.

6.2 Limitations

Due to platform incompatibility and insufficient probe coverage, the effort at external validation on GSE14520 failed, and the model was only trained and assessed using TCGA-LIHC data. Rather of using a held-out test set, the Graph Transformer was trained on the entire patient cohort, which could result in overly optimistic performance estimations. DNA methylation information, copy number variation, and somatic mutation profiles are not included in the current model, and histological grade information was not available for TCGA-LIHC patients.

6.3 Future Work

The 56-gene signature’s generalizability would be established by independent external validation on RNA-seq-based HCC cohorts like LIRI-JP (ICGC). The contribution of each architectural element would be explicitly demonstrated by an ablation research that contrasted the Graph Transformer with LASSO-Cox alone, a flat multilayer perceptron, and a standard GCN.

The most informative gene-gene interaction axes and patient similarity linkages for risk prediction would be revealed mechanistically by extracting the learnt attention weight matrices from the Graph Transformer. The molecular representation would be further enhanced by incorporating somatic mutation data in chromatin-related genes like ARID1A and CTNNB1, as well as DNA methylation data from TCGA-LIHC.

From a therapeutic standpoint, the results lend credence to the creation of epigenetic biomarker panels for early-stage HCC patient stratification in HDAC, EZH2, and PRMT5 inhibitor clinical studies.

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Complete LASSO-Cox Gene Signature

Table 1: Complete 56-gene LASSO-Cox signature

Gene	Coefficient	Family	Direction
KANSL1	+0.860	HMT	Risk
EZH2	+0.541	HMT	Risk
PRMT5	+0.434	PRMT	Risk
RUVBL1	+0.441	HMT	Risk
HDAC2	+0.420	HDAC	Risk
EP400	+0.445	HMT	Risk
KMT2D	-0.699	HMT	Protective
ING3	-0.514	HAT	Protective
SETD1B	-0.408	HMT	Protective
SUZ12	-0.372	HMT	Protective

Table 2: Complete 56-gene LASSO-Cox signature

Gene	Coefficient	Family	Direction
KDM1A	+0.335	KDM	Risk
HDAC4	+0.332	HDAC	Risk
HDAC1	+0.331	HDAC	Risk
RNF2	+0.324	CBX	Risk
NCOR2	−0.319	HDAC	Protective
USP7	−0.321	HMT	Protective
SIN3A	−0.311	HDAC	Protective
RING1	−0.291	CBX	Protective
SAP130	−0.276	HDAC	Protective
TRRAP	−0.253	HAT	Protective
KAT2A	−0.189	HAT	Protective
NCOA6	+0.283	HAT	Risk
OGT	−0.178	HMT	Protective
TAF7	−0.176	HAT	Protective
MORF4L2	−0.186	HMT	Protective
TP53	−0.192	HMT	Protective
KAT2B	+0.276	HAT	Risk

Table 3: Complete 56-gene LASSO-Cox signature

Gene	Coefficient	Family	Direction
KMT2C	+0.355	HMT	Risk
NCOR1	+0.134	HDAC	Risk
HDAC3	+0.136	HDAC	Risk
RCOR1	+0.138	KDM	Risk
EHMT2	+0.010	HMT	Risk
EP300	+0.100	HAT	Risk
PHF20	+0.100	HMT	Risk
CREBBP	+0.037	HAT	Risk
SAP30	+0.164	HDAC	Risk
TAF10	+0.206	HAT	Risk
ASH2L	−0.062	HMT	Protective
ATM	−0.072	HMT	Protective
BRCA1	−0.069	HMT	Protective
CDK1	−0.057	HMT	Protective
KAT5	−0.206	HAT	Protective
KAT8	+0.047	HAT	Risk
KDM6A	+0.032	KDM	Risk
KMT2A	−0.083	HMT	Protective
MAX	−0.023	HMT	Protective
MCRS1	−0.096	HMT	Protective
MEAF6	+0.042	HMT	Risk
RBBP5	−0.092	HMT	Protective
SIRT1	+0.073	SIRT	Risk
SUV39H1	+0.054	HMT	Risk
TADA3	−0.084	HAT	Protective
TAF1	−0.042	HAT	Protective
YEATS4	−0.109	HMT	Protective
SETD2	−0.026	HMT	Protective
ATF2	+0.000	HMT	Minimal

Inter-Family Interaction Network Statistics

Table 4: Complete interaction statistics (expanded 10-family network)

Family A	Family B	Type	Edges	Mean Conf.	Total Conf.	Max Conf.
HMT	HMT	Intra	128	851.6	109,010	999
HAT	HDAC	Inter	98	853.2	83,618	999
KDM	KDM	Intra	90	806.6	72,598	988
HMT	KDM	Inter	80	833.8	66,700	999
HAT	HAT	Intra	52	893.5	46,460	999
HDAC	HMT	Inter	50	842.1	42,104	999
HAT	HMT	Inter	48	825.5	39,626	995
HDAC	HDAC	Intra	44	840.1	36,964	999
PRMT	PRMT	Intra	12	892.0	10,704	993
HAT	PRMT	Inter	8	842.3	6,738	994
HDAC	PRMT	Inter	8	822.5	6,580	907
HMT	SIRT	Inter	6	905.3	5,432	942
SIRT	SIRT	Intra	6	866.0	5,196	956
KDM	SIRT	Inter	4	844.5	3,378	915
PRMT	SIRT	Inter	2	742.0	1,484	742
KDM	PRMT	Inter	2	727.0	1,454	727